torch.functional.tensordot#

https://pytorch.org/docs/stable/generated/torch.functional.tensordot.html

Description:

Returns a contraction of a and b over multiple dimensions. tensordot implements a generalized matrix product. a (Tensor) – Left tensor to contract b (Tensor) – Right tensor to contract dims (int or Tuple[List[int], List[int]] or List[List[int]] containing two lists or Tensor) – number of dimensions to contract or explicit lists of dimensions for a and b respectively

Function Signature

```
torch.functional.tensordot(a, b, dims=2, out=None)[source]#
Parameters
```

Code Examples

```
>>> a = torch.arange(60.).reshape(3, 4, 5)
>>> b = torch.arange(24.).reshape(4, 3, 2)
>>> torch.tensordot(a, b, dims=([1, 0], [0, 1]))
tensor([[4400., 4730.],
        [4532., 4874.],
        [4664., 5018.],
        [4796., 5162.],
        [4928., 5306.]])

>>> a = torch.randn(3, 4, 5, device='cuda')
>>> b = torch.randn(4, 5, 6, device='cuda')
>>> c = torch.tensordot(a, b, dims=2).cpu()
tensor([[ 8.3504, -2.5436,  6.2922,  2.7556, -1.0732,  3.2741],
        [ 3.3161,  0.0704,  5.0187, -0.4079, -4.3126,  4.8744],
        [ 0.8223,  3.9445,  3.2168, -0.2400,  3.4117,  1.7780]])

>>> a = torch.randn(3, 5, 4, 6)
>>> b = torch.randn(6, 4, 5, 3)
>>> torch.tensordot(a, b, dims=([2, 1, 3], [1, 2, 0]))
tensor([[  7.7193,  -2.4867, -10.3204],
        [  1.5513, -14.4737,  -6.5113],
        [ -0.2850,   4.2573,  -3.5997]])
```

Detailed Documentation

Documentation

Returns a contraction of a and b over multiple dimensions.

tensordot implements a generalized matrix product.

a (Tensor) – Left tensor to contract

b (Tensor) – Right tensor to contract

dims (int or Tuple[List[int], List[int]] or List[List[int]] containing two lists or Tensor) – number of dimensions to contract or explicit lists of dimensions for a and b respectively

When called with a non-negative integer argument dims = ddd, and the number of dimensions of a and b is mmm and nnn, respectively, tensordot() computes

When called with dims of the list form, the given dimensions will be contracted in place of the last ddd of a and the first ddd of bbb. The sizes in these dimensions must match, but tensordot() will deal with broadcasted dimensions.